



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0046



Agricultural Machinery – Forage Chopper JF ENS JF50MAX CX-JF50 MAXXIUM S3 FORAGE CHOPPER

Test Report Number	2022-0046
Date of Issuance	February 15, 2023
Valid Until	February 15, 2028
Agricultural Machinery	Forage Chopper
Type	Semi-precision cut, Flywheel cutterhead, Blow-up
Brand	JF
Model	ENS JF50MAX CX-JF50 MAXXIUM S3 FORAGE CHOPPER
Test Requested by	TRACTORCO COMPANY INC. Sitio Mambukog, Brgy. Zone 16 Talisay City, Negros Occidental, Philippines

SUMMARY

Performance Requirements	Requirement as per PAES 218:2004	AMTEC Test
Cutting Length Setting	At least three ^a	Two
Variation of Cut, %, maximum	5.0 ^a	12.90
Quality of Output	Clear cut chopped material	Clear cut chopped material
Noise Level, dB(A)	96 ^b	102.4

^a Requirement for precision and semi-precision cut.

^b Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English, Portuguese
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Provided for the Assembly and Installation
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	JF MAQUINAS AGRICOLAS LTDA Rua Santa Terezinha, 921 Jardim Guarujá, Itapira/SP - Brazil
4. Prime mover used	35.06 kW MASSEY FERGUSON MF240 Four-wheel tractor
5. Number of operators	Three (one tractor operator, and two feeders/baggers)
6. Loading of input	Feeding table is accessible to the feeding operators. Input materials were manually loaded into the feeding table.
7. Cleaning of parts	All parts are accessible for cleaning.
8. Adjusting and repairing of parts	Chopping length can be adjusted. Locally fabricated parts can be repaired
9. Collecting of output	Chopped materials were manually collected after the test operation.
10. Transporting the machine	The machine is mounted into the four- wheel tractor through the three-point linkage. Transporting the machine could be done by lifting the three-point linkage of the tractor.
11. Safety features	Power transmission system from the PTO to Gearbox 1 is provided with cover.
12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	None
13. Knives shall be made of AISI 1080 to AISI 1085 or its ISO equivalent.	Chopping blade material was not verified.

OBSERVATIONS (Cont'n)

14. Steel bars and heavy-duty mild steel shall be generally used for the manufacture of the different components of the forage chopper.	Steel materials were used in the fabrication of the machine.
15. Bolts and screws to be used shall conform with the requirements of PAES 311 and PAES 313.	It was not verified if the bolts and screws used conformed to PAES 311 and 312.
16. Cutting knives shall be accessible.	The cutting knives are accessible by opening the chopping chamber.
17. Knives shall be individually adjustable and replaceable and can be easily sharpened.	The cutting knives are bolted and can be easily adjusted, replaced and sharpened.
18. There should be provision for adjusting the clearance of flywheel/cylinder and stationary knife.	Provided
19. For random fed chopper, the length of the feeding table shall not be less than the arm length.	Not applicable
20. The base of the forage chopper shall be rigid, and all components shall be dynamically balanced for stable running with low noise level.	Stable running operation was observed during the test operation. However, it was not verified if all components are dynamically balanced.
21. The forage chopper shall be free from manufacturing defects that may be detrimental to its operation.	No manufacturing defects were observed during the test operation.
22. Any uncoated metallic surfaces shall be free from rust and shall be painted properly.	All parts are free from rust and properly painted.
23. The forage chopper shall be free from sharp edges and surfaces, other than the cutting mechanism, that may injure the operator.	Other than the cutting mechanism, no sharp edges and surfaces were observed that may injure the operator during the test operation.
24. Belt cover or guard and provisions for belt tightening and adjustments shall be provided.	Not applicable
25. Mechanism for immediate load disengagement between the prime mover and forage chopper shall be provided to protect the prime mover from overloading and the operator in case of accident.	A steel rod was used to engage/disengage the PTO shaft to the chopping blade assembly.
26. Other remarks	There were two cutting length settings (#5mm and #13mm) available in the machine. However, only the setting #13mm was used during the test operation.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 3, 2022
Location of Test	Sitio Mambukog, Brgy. Zone 16 Talisay City, Negros Occidental 10° 45' 10.00'' N, 122° 57' 56.00'' E
Items Tested	Operator's Manual, Noise Level, Fuel Consumption Rate, Input Capacity, Chopping Efficiency, Speed of Components, Number of Operators, and Quality of Output

METHODS OF TEST

The forage chopper was tested based on the PAES 219:2004 – Agricultural Machinery
– Forage Chopper – Methods of Test.

DESCRIPTION OF THE MACHINE

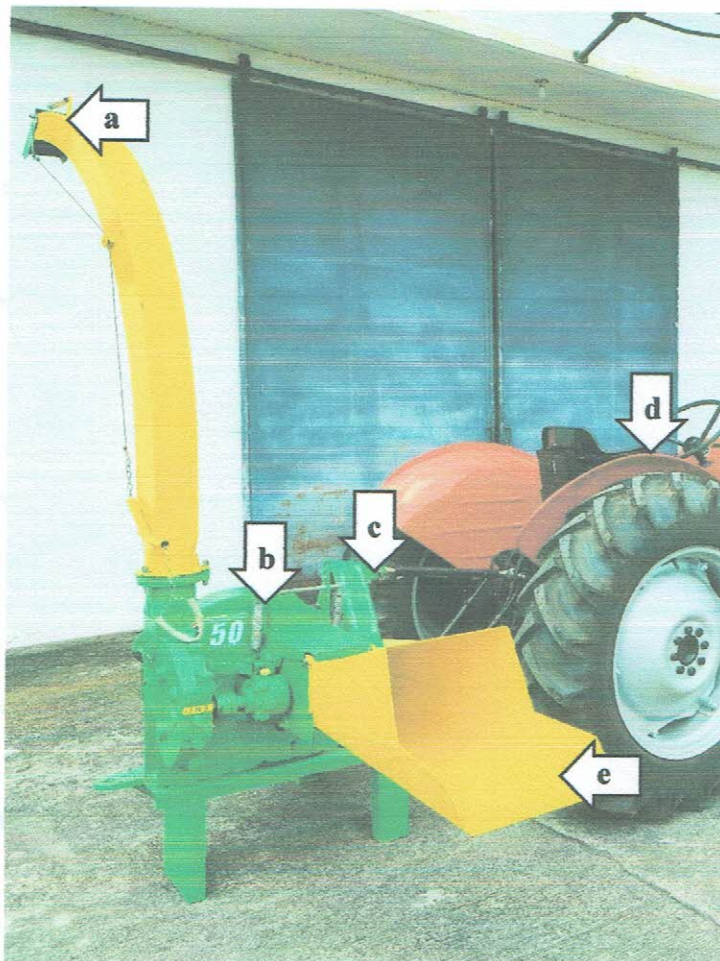


Figure 1. Parts of the JF ENS JF50MAX CX-JF50 MAXXIUM S3 FORAGE CHOPPER Forage Chopper powered by 35.06 kW MASSEY FERGUSON MF240 Four-wheel tractor include the following: (a) Output chute, (b) Chopping assembly, (c) Hitch point, (d) Prime mover, and (e) Input chute.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Main structure		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	4430
1.1.2 Width	ND	4830
1.1.3 Height	ND	2560
1.2 Forage chopper assembly		
1.2.1 Length	2698	2650
1.2.2 Width	2052	2050
1.2.3 Height	1620	2560
1.3 Weight, without prime mover, kg	ND	NM
2 Chopping assembly		
2.1 Type	ND	Cylinder cutterhead
2.2 Diameter, mm	ND	520
2.3 Cutting device		
2.3.1 Type	ND	Cutterhead knife
2.3.2 Dimensions, L × W × t, mm	ND	210 × 79.68 × 8.44
2.3.3 Number of knives	ND	4 Rotating, 1 stationary
2.3.4 Means of attachment	ND	Bolts and nuts
2.3.5 Materials	ND	ND
3 Feeding table		
3.1 Dimension, L × W, mm	ND	985 × 400
3.2 Height from the ground, mm	ND	655
3.3 Material	ND	ND
4 Feeding inlet		
4.1 Dimension, L × H, mm	ND	180 × 180
5 Output chute		
5.1 Dimensions, L × W, mm	ND	125 × 100
5.2 Height from the ground, mm	ND	2440
6 Main frame		
6.1 Material	ND	Channel bars
6.2 Dimension, L × W × t, mm	ND	900 × 945 × 715
6.3 Channel bar thickness, mm	ND	7.10

^a The data were obtained from the operator's manual provided by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
7	Tires	NA	NA
8	Power transmission system		
8.1	Four-wheel tractor PTO to gearbox 1	ND	Universal joint and cardan shaft
8.1.1	Gearbox 1 to chopper	ND	Direct-coupled
8.1.2	Chopper to gearbox 2	ND	Direct-coupled
8.1.3	Gearbox 2 to feed rollers	ND	Cardan shaft
9	Four-wheel tractor		
9.1	Brand	MASSEY FERGUSON	MASSEY FERGUSON
9.2	Model	MF240	MF240
9.3	Serial number	ND	ND
9.4	Type	ND	Four-stroke, 3-cylinder, naturally aspirated diesel engine
9.5	Rated power, kW	35.06	NM
9.6	Rated speed, rpm	540	NM
9.7	Cooling system	Water-cooled	Water-cooled
9.8	Starting system	Key starter	Key starter
9.9	Weight, kg	ND	NM

^a The data were obtained from the tractors nameplate.

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Conditions of test material		
1.1	Test material	Napier grass	
1.2	Variety	ND	
1.3	Source	Talisay, Negros Occidental	
1.4	Moisture content, % _{wb}	85.48	
1.5	Average length of plant, mm	2051	
1.6	Average diameter of plant, mm	12.1	
2	Performance test		
2.1	Weight of input, kg	224.0	
2.2	Input capacity, kg/h	3591.98	
2.3	Chopping efficiency, %	100.0	
2.4	Speed of components, rpm	Without Load	With Load
2.4.1	Engine	623	608
2.4.2	Feeding shaft (Top roller)	220	213
2.4.3	Chopping shaft	NM	NM
2.5	Noise level, dB(A)		
2.5.1	Feeding operator	96.5	102.4
2.5.2	Bagger	98.1	99.6
2.6	Fuel consumption rate, L/h	3.30	
2.7	Average length of cut, mm	12.35	
2.8	Coefficient of variation of cut, %	12.90	

Note: ND – No Data, NM – Not Measured

OBSERVATIONS

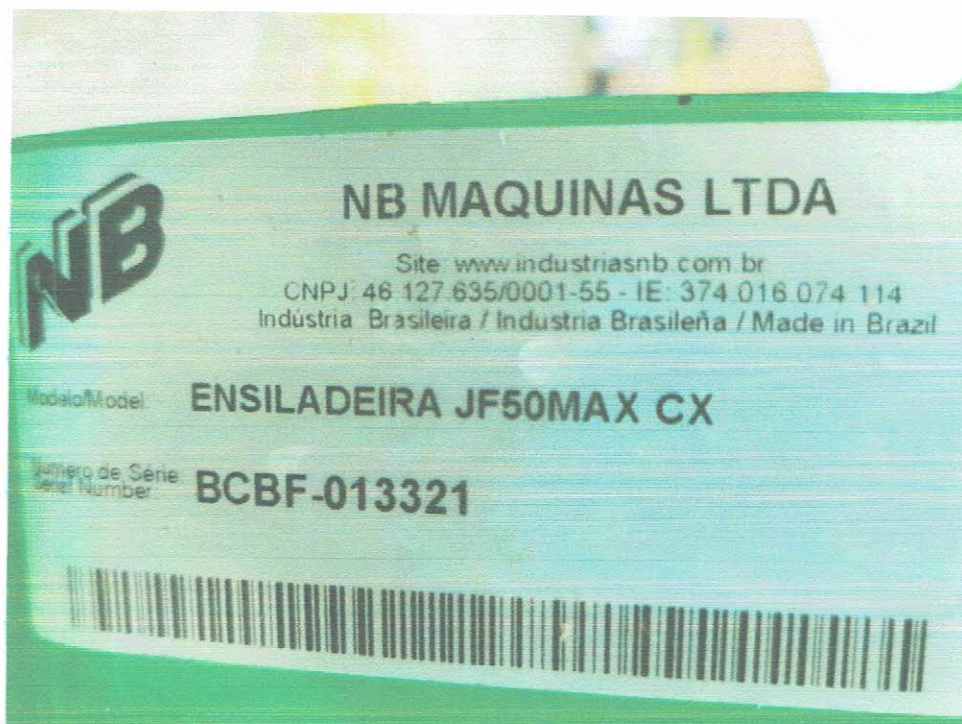


Figure 2. Nameplate of the JF ENS JF50MAX CX-JF50 MAXXIUM S3 FORAGE CHOPPER Forage Chopper.

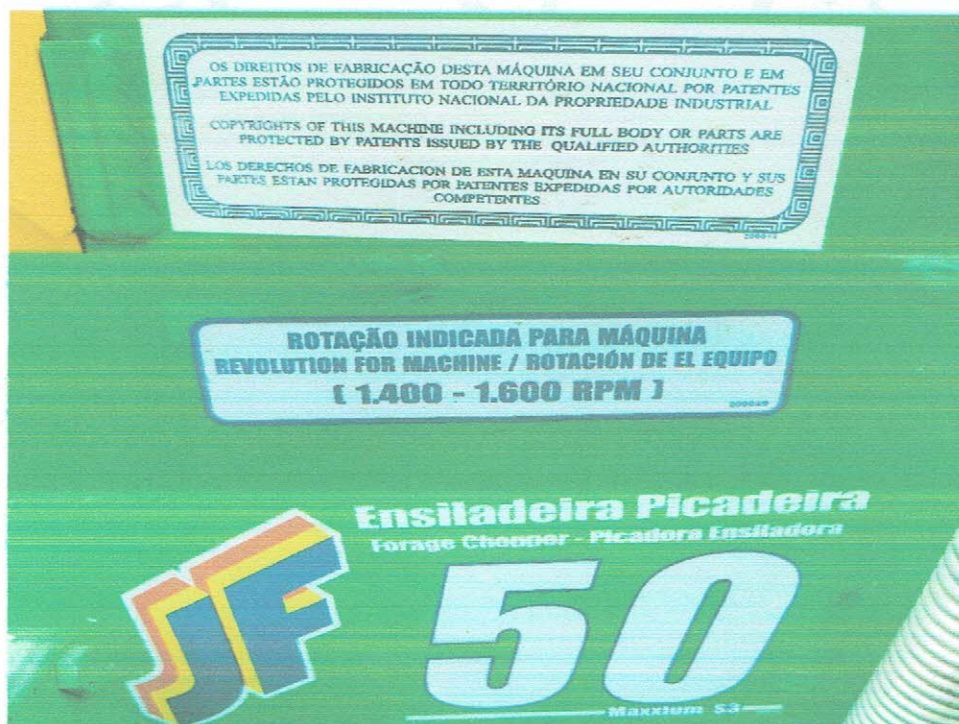


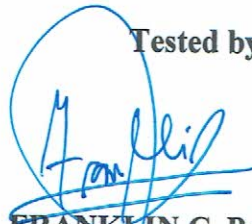
Figure 3. Markings on the JF ENS JF50MAX CX-JF50 MAXXIUM S3 FORAGE CHOPPER Forage Chopper.

OBSERVATIONS (Cont'n)



Figure 4. Quality of output.

Tested by:



FRANKLIN C. PAJARON

Test Engineer

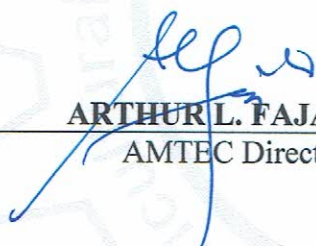
Verified by:



MARIE JEHOA B. REYES

Test Engineer

Approved for Release:



ARTHUR L. FAJARDO

AMTEC Director

FEB 15 2023

Date

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